

Sahrdaya College of Engineering and
Technology, Kodakara

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Second Semester B.Tech Degree Examination June 2022
(2019 Scheme)

Course Code : MAT102
Vector Calculus, Differential Equations and Transforms

PART A

1. Find the directional derivatives of $f(x,y) = x^2 - 3xy + y^2$ at the point $P(2,1)$ in the direction of $\vec{a} = \frac{1}{3}\hat{i} + \frac{2}{3}\hat{j}$

$$D_u f(x_0, y_0) = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2$$

$$f_x = 2x - 3y, \quad f_x(2,1) = 2 \times 2 - 3 \times 1 = 1$$

$$f_y = -3x + 2y, \quad f_y(2,1) = -3 \times 2 + 2 \times 1 = -4$$

$$u_1 = \frac{1}{\sqrt{5}}, \quad u_2 = \frac{2}{\sqrt{5}}$$

$$\left| \vec{a} = \frac{\frac{1}{3}\hat{i} + \frac{2}{3}\hat{j}}{\sqrt{\frac{1}{9} + \frac{4}{9}}} \right|$$

$$D_u f = 1 \times \frac{1}{\sqrt{5}} + (-4) \times \frac{2}{\sqrt{5}} = \underline{\underline{\frac{-7}{\sqrt{5}}}}$$

2. Evaluate $\int_C 3xy \, dy$ where C is the line segment joining $(0,0)$ and $(1,2)$.

Equation of line segment joining $(0,0)$ and $(1,2)$

$$\frac{x-0}{1-0} = \frac{y-0}{2-0} \quad \therefore y = 2x, \quad dy = 2 \, dx$$

$$\int_C 3xy \, dy = 3 \int_0^1 x \cdot 2x \cdot 2 \, dx = 12 \int_0^1 x^2 \, dx$$

$$= 12 \left[\frac{x^3}{3} \right]_0^1$$

$$= 4(1-0) = \underline{\underline{4}}$$

3. Determine the sources and sinks of the vector field $\vec{F}(x,y,z) = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$

$$\nabla \cdot \vec{F} = 2x + 2y + 2z$$

$$= 2(x+y+z)$$

If $2(x+y+z) > 0$ represents source,
 $2(x+y+z) < 0$ represents sink.

4. Using divergence theorem, evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ where $\vec{F} = 2x\hat{i} + 4y\hat{j} - 3z\hat{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = 1$.

Divergence theorem is $\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_C (\nabla \cdot \vec{F}) \, dv$

$$\nabla \cdot \vec{F} = 2\hat{i} + 4\hat{j} - 3\hat{k}$$

$$\nabla \cdot \vec{F} = 2 + 4 - 3 = 3$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_C 3 \, dv$$

$$= 3 \times \text{volume of sphere}$$

$$= 3 \cdot \frac{4}{3} \pi r^3 = 4\pi \cdot 1^3 = \underline{\underline{4\pi}}$$

5. Solve the initial value problem $y'' + 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 2$.

$$\text{AE is } m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

$$\therefore m = -2, -3$$

$$y = c_1 e^{-2x} + c_2 e^{-3x}$$

$$y(0) = 1$$

$$\therefore 1 = c_1 + c_2 \quad \text{--- (1)}$$

$$y' = -2c_1 e^{-2x} - 3c_2 e^{-3x}$$

$$\text{Given } y'(0) = 2.$$

$$y'(0) = -2c_1 - 3c_2 = 2. \quad \text{--- (2)}$$

$$\text{From (1), } c_2 = 1 - c_1$$

$$\text{Substituting in (2), } 2 = -2c_1 - 3(1 - c_1)$$

$$2 = -2c_1 - 3 + 3c_1$$

$$\underline{5 = c_1} \quad \therefore c_2 = 1 - 5 = \underline{-4}$$

$$\text{Thus } \underline{y = 5e^{-2x} - 4e^{-3x}} \text{ is the solution.}$$

6. Solve $y''' - y' = 0$

$$\text{AE is } m^3 - m = 0$$

$$m(m^2 - 1) = 0$$

$$m = 0, \quad m = \pm 1$$

$$\therefore \text{solution is } y = c_1 e^{0x} + c_2 e^x + c_3 e^{-x}$$

$$\underline{y = c_1 + c_2 e^x + c_3 e^{-x}}$$

7. Find the Laplace transform of $(\sin t + \cos t)^2$.

$$L[(\sin t + \cos t)^2] = L[\sin^2 t + 2\sin t \cos t + \cos^2 t]$$

$$= L[1 + 2\sin t \cos t] = L[1 + \sin 2t]$$

$$= L[1] + L[\sin 2t] = \underline{\underline{\frac{1}{s} + \frac{2}{s^2 + 4}}}$$

10. Find the Fourier Sine integral of

$$f(x) = \begin{cases} \sin x & , \text{ if } 0 < x < \pi \\ 0 & , \text{ if } x > \pi \end{cases}$$

Fourier sine integral is

$$f(x) = \int_0^{\infty} B(\omega) \sin \omega x \, d\omega \quad \text{and}$$

$$B(\omega) = \frac{2}{\pi} \int_0^{\infty} f(x) \sin \omega x \, dx.$$

$$= \frac{2}{\pi} \int_0^{\pi} \sin x \sin \omega x \, dx$$

$$= \frac{1}{\pi} \int_0^{\pi} 2 \sin x \sin \omega x \, dx \quad \left| \begin{array}{l} 2 \sin A \sin B \\ = \cos(A-B) \\ - \cos(A+B) \end{array} \right.$$

$$= \frac{1}{\pi} \int_0^{\pi} \cos(x-\omega x) - \cos(x+\omega x) \, dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \cos(1-\omega)x - \cos(1+\omega)x \, dx$$

$$= \frac{1}{\pi} \left[\frac{\sin(1-\omega)x}{1-\omega} - \frac{\sin(1+\omega)x}{1+\omega} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\sin(\pi - \pi\omega)}{1-\omega} - \frac{\sin(\pi + \pi\omega)}{1+\omega} \right]$$

$$= \frac{1}{\pi} \left[\frac{\sin \pi\omega}{1-\omega} - \frac{-\sin \pi\omega}{1+\omega} \right]$$

$$= \frac{\sin \pi\omega}{\pi} \left[\frac{1}{1-\omega} + \frac{1}{1+\omega} \right]$$

$$= \frac{\sin \pi\omega}{\pi} \left(\frac{1+\omega + 1-\omega}{1-\omega^2} \right)$$

$$= \frac{2 \sin \pi\omega}{\pi(1-\omega^2)}$$

Part B

Module I

- 11 a) Find the parametric equation of the tangent line to the curve $\vec{r}(t) = 2\cos\pi t \hat{i} + 2\sin\pi t \hat{j} + 6t \hat{k}$ at the point $t = \frac{1}{3}$

sol: Equation of tangent line to $\vec{r}(t)$ at t_0 is,

$$\vec{r}(t) = \vec{r}(t_0) + t \vec{r}'(t_0) \text{ ——— (1)}$$

Here, $t_0 = \frac{1}{3}$ $\therefore$ (1) $\Rightarrow \vec{r}(t) = \vec{r}(\frac{1}{3}) + t \vec{r}'(\frac{1}{3})$ — (2)

To find $\vec{r}(\frac{1}{3})$:

Given: $\vec{r}(t) = 2\cos\pi t \hat{i} + 2\sin\pi t \hat{j} + 6t \hat{k}$ — (A)

$$\therefore \vec{r}(\frac{1}{3}) = 2\cos\frac{\pi}{3} \hat{i} + 2\sin\frac{\pi}{3} \hat{j} + \frac{6}{3} \hat{k}$$

$$= 2 \cdot \frac{1}{2} \hat{i} + 2 \cdot \frac{\sqrt{3}}{2} \hat{j} + 2 \hat{k}$$

$$= \hat{i} + \sqrt{3} \hat{j} + 2 \hat{k} \text{ — (3)}$$

To find $\vec{r}'(\frac{1}{3})$

Differentiating (A); $\vec{r}'(t) = -2\sin\pi t \hat{i} \cdot \pi + 2\pi \cos\pi t \hat{j} + 6 \hat{k}$

$$\Rightarrow \vec{r}'(\frac{1}{3}) = -2\sin\frac{\pi}{3} \hat{i} \cdot \pi + 2\pi \cos\frac{\pi}{3} \hat{j} + 6 \hat{k}$$

$$= -2 \cdot \frac{\sqrt{3}}{2} \hat{i} \cdot \pi + 2\pi \cdot \frac{1}{2} \hat{j} + 6 \hat{k}$$

$$= -\sqrt{3}\pi \hat{i} + \pi \hat{j} + 6 \hat{k} \text{ — (4)}$$

Substituting ③ + ④ in ②,

$$\begin{aligned}\vec{r}(t) &= [\hat{i} + \sqrt{3}\hat{j} + 2\hat{k}] + t[-\sqrt{3}\pi\hat{i} + \pi\hat{j} + 6\hat{k}] \\ &= \hat{i}[1 - \sqrt{3}\pi t] + \hat{j}[\sqrt{3} + \pi t] + \hat{k}[2 + 6t]\end{aligned}$$

$\therefore$ Parametric equations of the tangent line are

$$\underline{x = 1 - \sqrt{3}\pi t, \quad y = \sqrt{3} + \pi t, \quad z = 2 + 6t}$$

11) ⑤ Show that the vector field $\vec{F}(x,y) = 2xy^3\hat{i} + 3y^2x^2\hat{j}$ is conservative and find ϕ such that $\vec{F} = \nabla\phi$.

Hence evaluate $\int_{(2,-2)}^{(2,0)} 2xy^3 dx + 3y^2x^2 dy$.

sol: To show that $\vec{F}$ is conservative;

$$\begin{aligned}\nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy^3 & 3y^2x^2 & 0 \end{vmatrix} = \hat{i}(0-0) - \hat{j}(0-0) \\ &\quad + \hat{k}(6y^2x - 6xy^2) \\ &= 0\end{aligned}$$

$\therefore \vec{F}$ is conservative as $\nabla \times \vec{F} = 0$

To find ϕ

$$\vec{F} = \nabla \phi$$

$$\Rightarrow 2xy^3 \hat{i} + 3y^2 x^2 \hat{j} = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y}$$

$$\Rightarrow \left\{ \frac{\partial \phi}{\partial x} = 2xy^3 \Rightarrow \phi = 2 \frac{x^2}{2} y^3 + \text{fn not without } x \right.$$

$$\left\{ \frac{\partial \phi}{\partial y} = 3y^2 x^2 \Rightarrow \phi = \frac{3y^3}{3} x^2 + \text{fn without } y \right.$$

$$\therefore \underline{\underline{\phi = x^2 y^3 + K}}$$

To evaluate $\int_{(2,-2)}^{(-2,0)} 2xy^3 dx + 3y^2 x^2 dy$

$$\int_{(2,-2)}^{(-2,0)} 2xy^3 dx + 3y^2 x^2 dy = \phi(-2,0) - \phi(2,-2)$$

$$= (0 + K) - (-32 + K)$$

$$= K + 32 - K$$

$$= \underline{\underline{32}}$$

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① Find the position and velocity vectors of the particle.

Given $\vec{a}(t) = (t+1)^{-2} \hat{j} - e^{-2t} \hat{k}$, $\vec{v}(0) = 3\hat{i} - \hat{j}$, $\vec{r}(0) = \hat{k}$

Sol: $\vec{a}(t) = (t+1)^{-2} \hat{j} - e^{-2t} \hat{k}$

Integrating;

$$\begin{aligned} \vec{v}(t) &= \int \vec{a}(t) dt = \int ((t+1)^{-2} \hat{j} - e^{-2t} \hat{k}) dt \\ &= \frac{(t+1)^{-1}}{-1} \hat{j} - \frac{e^{-2t}}{-2} \hat{k} + C \end{aligned}$$

i.e. $\vec{v}(t) = -\frac{1}{t+1} \hat{j} + \frac{e^{-2t}}{2} \hat{k} + C$ — (1)

Given, $\vec{v}(0) = 3\hat{i} - \hat{j}$

putting $t=0$ in (1),

$$\begin{aligned} \vec{v}(0) &= -1 \hat{j} + \frac{1}{2} \hat{k} + C \\ \left. \begin{aligned} \vec{v}(0) &= 3\hat{i} - \hat{j} \\ \vec{v}(0) &= -1 \hat{j} + \frac{1}{2} \hat{k} + C \end{aligned} \right\} \begin{aligned} &\rightarrow 3\hat{i} - \hat{j} = -\hat{j} + \frac{1}{2} \hat{k} + C \\ &\rightarrow C = 3\hat{i} - \frac{1}{2} \hat{k} \end{aligned} \end{aligned}$$

$\therefore$ (1) becomes,

$$\vec{v}(t) = -\frac{1}{t+1} \hat{j} + \frac{e^{-2t}}{2} \hat{k} + 3\hat{i} - \frac{1}{2} \hat{k}$$

$$\Rightarrow \underline{\underline{\vec{v}(t) = 3\hat{i} - \frac{1}{t+1} \hat{j} + \frac{e^{-2t}-1}{2} \hat{k}}} \quad \text{is the required velocity vector.} \quad (2)$$

Integrating velocity vector $\vec{v}(t)$, we get position vector.

Integrating (2),

$$\vec{r}(t) = \int \vec{v}(t) dt = 3t \hat{i} - \log(t+1) \hat{j} + \left(\frac{e^{-2t}}{2 \times -2} - \frac{1}{2} t \right) \hat{k} + C_1 \quad \text{--- (3)}$$

Given: $\vec{r}(0) = \hat{k}$

Put $t=0$ in (3); we get

$$\vec{r}(0) = 0 \hat{i} - 0 \hat{j} - \frac{1}{4} \hat{k} + C_1$$

$$\hat{k} = -\frac{1}{4} \hat{k} + C_1$$

$$C_1 = \frac{5}{4} \hat{k}$$

Substituting $C_1 = \frac{5}{4} \hat{k}$ in (3),

$$\vec{r}(t) = 3t \hat{i} - \log(t+1) \hat{j} + \left(\frac{e^{-2t}}{-4} - \frac{1}{2} t \right) \hat{k} + \frac{5}{4} \hat{k}$$

$$\text{ie } \vec{r}(t) = 3t \hat{i} - \log(t+1) \hat{j} + \left(\frac{e^{-2t}}{-4} - \frac{t}{2} + \frac{5}{4} \right) \hat{k}$$

which is the required position vector of the particle.

6) If $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$ and let $\vec{F}(\vec{r}) = f(r) \vec{r}$,
prove that $\text{div } \vec{F} = 3f(r) + \vec{r} f'(r)$

$$\begin{aligned} \therefore \text{div } \vec{F} &= \text{div} [f(r) \vec{r}] \\ &= \text{div} [f(r) (x \hat{i} + y \hat{j} + z \hat{k})] \end{aligned}$$

$$\operatorname{div} \vec{F} = \frac{\partial}{\partial x} [f(r)x] + \frac{\partial}{\partial y} [f(r)y] + \frac{\partial}{\partial z} [f(r)z] \quad \text{--- (1)}$$

To find $\frac{\partial}{\partial x} [f(r)x]$

$$\begin{aligned} \frac{\partial}{\partial x} [f(r)x] &= f(r) \cdot 1 + x \cdot f'(r) \cdot \frac{\partial r}{\partial x} \\ &= f(r) + x f'(r) \cdot \frac{x}{r} \end{aligned}$$

$$= f(r) + \frac{x^2}{r} f'(r)$$

Similarly ; $\frac{\partial}{\partial y} [f(r)y] = f(r) + \frac{y^2}{r} f'(r)$ and

$$\frac{\partial}{\partial z} [f(r)z] = f(r) + \frac{z^2}{r} f'(r)$$

Substituting all in (1) ;

$$\operatorname{div} \vec{F} = f(r) + \frac{x^2}{r} f'(r) + f(r) + \frac{y^2}{r} f'(r) + f(r) + \frac{z^2}{r} f'(r)$$

$$= 3f(r) + \frac{f'(r)}{r} (x^2 + y^2 + z^2)$$

$$= 3f(r) + \frac{f'(r)}{r} \cdot r^2$$

$$= 3f(r) + r f'(r)$$

$$\left\{ \begin{array}{l} r = |\vec{r}| \\ \Rightarrow r = \sqrt{x^2 + y^2 + z^2} \\ \Rightarrow r^2 = x^2 + y^2 + z^2 \end{array} \right.$$

$$\begin{aligned}
 W &= \int_{\theta=0}^{\pi} \sin \theta d\theta \times \int_{r=0}^4 r^2 dr \\
 &= [-\cos \theta]_0^{\pi} \times \left[\frac{r^3}{3} \right]_0^4 = (-\cos \pi - \cos 0) \times \frac{4^3}{3} = \underline{\underline{\frac{128}{3}}}
 \end{aligned}$$

13. ⑥ Find mass of the lamina that is a portion of cone $z = \sqrt{x^2 + y^2}$ that lies between the planes $z=1$ and $z=3$, if the density is $\phi(x, y, z) = x^2 z$.

Sol. Mass = $\iint_{\sigma} \phi(x, y, z) ds = \iint_{\sigma} x^2 z ds$

Projecting to xy -plane;

$$\text{Mass} = \iint_R x^2 \sqrt{x^2 + y^2} \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dx dy \quad \text{--- (1)}$$

$$\begin{aligned}
 z &= \sqrt{x^2 + y^2} \implies \frac{\partial z}{\partial x} = \frac{1}{2\sqrt{x^2 + y^2}} \cdot 2x = \frac{x}{\sqrt{x^2 + y^2}} \\
 \frac{\partial z}{\partial y} &= \frac{y}{\sqrt{x^2 + y^2}}
 \end{aligned}$$

$$\therefore \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} = \sqrt{\frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2} + 1} = \sqrt{2}$$

i) ① becomes,

$$\text{Mass} = \iint_R x^2 \sqrt{x^2 + y^2} \sqrt{2} \, dx \, dy$$

$$= \int_{\theta=0}^{2\pi} \int_{r=1}^3 r^2 \cos^2 \theta \cdot r \cdot \sqrt{2} \cdot r \, dr \, d\theta$$

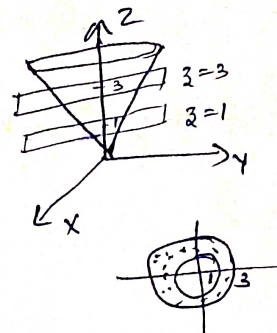
$$= \sqrt{2} \int_{\theta=0}^{2\pi} \cos^2 \theta \, d\theta \cdot \int_{r=1}^3 r^4 \, dr$$

$$= \sqrt{2} \int_{\theta=0}^{2\pi} \frac{1 + \cos 2\theta}{2} \, d\theta \cdot \left[\frac{r^5}{5} \right]_{r=1}^3$$

$$= \frac{\sqrt{2}}{2} \left[\theta + \frac{\sin 2\theta}{2} \right]_{\theta=0}^{2\pi} \cdot \left(\frac{3^5}{5} - \frac{1}{5} \right)$$

$$= \frac{\sqrt{2}}{2} \left[(2\pi + 0) - (0 + 0) \right] \cdot \left(\frac{3^5 - 1}{5} \right)$$

$$= \sqrt{2} \pi \left(\frac{242}{5} \right) \text{ units}$$



14a) Let σ be the portion of the surface $z = 1 - x^2 - y^2$ that lies above the xy -plane and σ is oriented upwards. Find the flux of the vector field $\vec{F}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$ across σ .

Q: Flux = $\iint_{\sigma} \vec{F} \cdot \vec{n} \, d\vec{s}$

$$= \iiint_V \nabla \cdot \vec{F} \, d\vec{v}$$

$$= \iiint_V 3 \, d\vec{v}$$

$$= 3 \int \int \int_{z=0}^{1-x^2-y^2} dz \, dA = 3 \int \int (1-x^2-y^2) \, dA$$

Converting to polar co-ordinates, $dA = r \, dr \, d\theta$

$$x^2 + y^2 = r^2$$

$$r: 0 \rightarrow 1, \theta: 0 \rightarrow 2\pi$$

$$\text{Flux} = 3 \int_{\theta=0}^{2\pi} \int_{r=0}^1 (1-r^2) r \, dr \, d\theta$$

$$= 3 \int_{\theta=0}^{2\pi} \left[\int_{r=0}^1 (r - r^3) \, dr \right] d\theta$$

$$= 3 \int_{\theta=0}^{2\pi} \left[\frac{r^2}{2} - \frac{r^4}{4} \right]_{r=0}^1 d\theta$$

$$= 3 \int_{\theta=0}^{2\pi} \left(\frac{1}{2} - \frac{1}{4} \right) d\theta$$

$$= 3 \times \frac{1}{4} \left[\theta \right]_0^{2\pi}$$

$$= \frac{3}{4} [2\pi - 0]$$

$$= \underline{\underline{\frac{3}{2} \pi}}$$

14) b) Use Stokes theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y, z) = z^2 \hat{i} + 3xz \hat{j} - y^3 \hat{k}$ and C is the circle $x^2 + y^2 = 1$ in the xy -plane with counter clockwise orientation looking down the positive z -axis.

sol: Stokes theorem: $\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \hat{n} \, dA$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & 3xz & -y^3 \end{vmatrix}$$

$$= \hat{i}(-3y^2 - 0) - \hat{j}(0 - 2z) + \hat{k}(3 - 0)$$

$$= -3y^2 \hat{i} + 2z \hat{j} + 3 \hat{k}$$

$$\hat{n} = \hat{k} \quad \therefore (\nabla \times \vec{F}) \cdot \hat{n} = 3$$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \iint_R 3 \, dA$$

$$= 3 \times \text{area of circle } x^2 + y^2 = 1$$

$$= 3 \times \pi (1)^2$$

$$= \underline{\underline{3\pi}}$$

Module III

15) a) Using the method of undetermined coefficients, solve
 $y'' - 4y = xe^x$

Ans: $y'' - 4y = xe^x$ — (1)

Solution of (1) is ; $y = C.F + P.I$ — (2)

To find C.F

$$y'' - 4y = 0 \Rightarrow (D^2 - 4)y = 0 \Rightarrow A.E : D^2 - 4 = 0$$
$$\Rightarrow D^2 = 4$$
$$\Rightarrow D = \pm 2$$

$$\therefore C.F = C_1 e^{2x} + C_2 e^{-2x} \text{ — (3)}$$

To find P.I : $RHS = xe^x$

Choose $y_p = (Ax + B)e^x$ — (4)

$$\Rightarrow y_p = Ax e^x + B e^x$$

$$y_p' = A[xe^x + e^x] + B e^x$$

$$y_p'' = A[xe^x + e^x + e^x] + B e^x$$

Substituting all in (1), we get

$$A [xe^x + 2e^x] + Be^x - 4 [Axe^x + Be^x] = xe^x$$

$$\Rightarrow e^x [Ax + 2A + B - 4Ax - 4B] = xe^x$$

$$\Rightarrow Ax + 2A - 3B - 4Ax = x$$

Equating coefficients of x on both sides,

$$A - 4A = 1 \Rightarrow -3A = 1 \Rightarrow \underline{\underline{A = -1/3}}$$

Equating constants on both sides,

$$2A - 3B = 0$$

$$\Rightarrow 2A = 3B$$

$$\Rightarrow B = \frac{2}{3}A$$

$$\Rightarrow B = \frac{2}{3} \times -\frac{1}{3}$$

$$\Rightarrow B = -\frac{2}{9}$$

Thus (4) becomes, $y_p = \left(-\frac{1}{3}x - \frac{2}{9}\right)e^x$

$$\text{ie } p.I = \left(-\frac{1}{3}x - \frac{2}{9}\right)e^x$$

Solution (2) becomes,

$$\underline{\underline{y = C_1 e^{2x} + C_2 e^{-2x} - \left(\frac{1}{3}x + \frac{2}{9}\right)e^x}}$$

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b) Using the method of variation of parameters,

solve: $y'' - 4y' + 5y = \frac{e^{2x}}{\sin x}$

Sol: Solution of given o.d.e is, $y = C.F + P.I$ — ①

To find C.F

$$y'' - 4y' + 5y = 0 \Rightarrow (D^2 - 4D + 5)y = 0$$

$$\Rightarrow A.E: D^2 - 4D + 5 = 0$$

$$\Rightarrow D = \frac{4 \pm \sqrt{16 - 20}}{2}$$

$$= \frac{4 \pm \sqrt{4}}{2} = \frac{4 \pm 2i}{2}$$

$$= 2 \pm i$$

$$\therefore C.F = e^{2x} (C_1 \cos x + C_2 \sin x) \text{ — ②}$$

To find P.I

$$P.I = -y_1 \int \frac{y_2 r}{w} dx + y_2 \int \frac{y_1 r}{w} dx \text{ — ③}$$

Here, $r = \frac{e^{2x}}{\sin x}$, $y_1 = e^{2x} \cos x$, $y_2 = e^{2x} \sin x$

$$w = \begin{vmatrix} e^{2x} \cos x & e^{2x} \sin x \\ e^{2x} \cdot \sin x + \cos x \cdot 2e^{2x} & e^{2x} \cos x + \sin x \cdot 2e^{2x} \end{vmatrix}$$

$$= e^{4x} \cos^2 x + e^{4x} 2 \cos x \sin x + e^{4x} \sin^2 x - e^{4x} 2 \sin x \cos x$$

$$= e^{4x} (\cos^2 x + \sin^2 x) = e^{4x}$$

Substituting all in (3),

$$P.I = -e^{2x} \cos x \int \frac{e^{2x} \sin x}{e^{4x}} \cdot \frac{e^{2x}}{\sin x} dx +$$

$$e^{2x} \sin x \int \frac{e^{2x} \cos x}{e^{4x}} \cdot \frac{e^{2x}}{\sin x} dx$$

$$= -e^{2x} \cos x \int dx + e^{2x} \sin x \int dx \cot x dx$$

$$= -e^{2x} \cos x \cdot x + e^{2x} \sin x \cdot \log(\sin x) \quad \text{--- (4)}$$

Substituting (2) & (4) in (1) we get the required solution.

16. a)

Solve the i.v.p by the method of undetermined coefficients $y'' + 4y = 8x^2$, $y(0) = -3$, $y'(0) = 0$

Sol:

$$A.E: D^2 + 4 = 0 \Rightarrow D^2 = -4 \Rightarrow D = \pm 2i$$

$$\therefore C.F = C_1 \cos 2x + C_2 \sin 2x \quad \text{or} \quad y_h = \underline{C_1 \cos 2x + C_2 \sin 2x} \quad \text{--- (1)}$$

To find P.I (y_p): Using undetermined coefficient method, choose $P.I = Ax^2 + Bx + C$

$$y_p = Ax^2 + Bx + C$$

$$y_p' = 2Ax + B$$

$$y_p'' = 2A$$

$$y'' + 4y = 8x^2 \Rightarrow 2A + 4(Ax^2 + Bx + C) = 8x^2$$

$$\Rightarrow 2A + 4Ax^2 + 4Bx + 4C = 8x^2$$

Equating coefficients of x^2 on both sides: $4A = 8$

$$\underline{\underline{A = 2}}$$

" " " " " " : $4B = 0$

$$\underline{\underline{B = 0}}$$

Equating constants on both sides; $2A + 4C = 0$

$$\Rightarrow 4 + 4C = 0$$

$$\Rightarrow 4 = -4C$$

$$\Rightarrow \underline{\underline{C = -1}}$$

$$\therefore \underline{\underline{y_p = 2x^2 - 1}} \quad \text{--- (2)}$$

Solution of given d.e is $y = y_h + y_p$. From (1) & (2)

$$y = C_1 \cos 2x + C_2 \sin 2x + 2x^2 - 1 \quad \text{--- (3)}$$

Given: $y(0) = -3$

Put $x=0$ in (3), then

$$-3 = C_1 + 0 + 0 - 1 \Rightarrow C_1 = -3 + 1 = \underline{\underline{-2}}$$

Also, given that $y'(0)=0$.

Differentiating (3);

$$y' = C_1 \cdot 2 \sin 2x + C_2 \cdot 2 \cos 2x + 4x$$

put $x=0$, then

$$0 = 0 + 2C_2 + 0 \Rightarrow \underline{\underline{C_2 = 0}}$$

Substituting $C_1 = -2$, $C_2 = 0$ in (3), we get

$$\underline{\underline{y = -2 \cos 2x + 2x^2 - 1}} \text{ which is the required solution.}$$

6b) Solve the i.v.p $x^2 y'' + 3xy' + y = 0$, $y(1) = -3$, $y'(1) = 1$

$$\text{Sol: } x^2 y'' + 3xy' + y = 0 \equiv x^2 y'' + axy' + by = 0 \Rightarrow \begin{matrix} a=3 \\ b=1 \end{matrix}$$

$$\text{A.E: } m^2 + (a-1)m + b = 0 \Rightarrow m^2 + 2m + 1 = 0 \\ \Rightarrow m = -1, -1$$

$$\therefore y = (C_1 + C_2 \log x) x^{-1} \quad \text{--- (1)}$$

Given: $y(1) = -3$; putting $x=1$ in (1),

$$y(1) = (C_1 + 0) 1^{-1} = -3$$

$$\Rightarrow \underline{\underline{C_1 = -3}}$$

Differentiating (1),

Module 4.

119 Using Laplace Transform solve $y'' + 5y' + 6y = e^{-t}$
 $y(0) = 0$ $y'(0) = 1$.

Ans. $y'' + 5y' + 6y = e^{-t}$

$$\mathcal{L}(y'' + 5y' + 6y) = \mathcal{L}(e^{-t})$$

$$\mathcal{L}(y'') + 5\mathcal{L}(y') + 6\mathcal{L}(y) = \mathcal{L}(e^{-t})$$

$$[s^2\mathcal{L}(y) - sy(0) - y'(0)] + 5[s\mathcal{L}(y) - y(0)] + 6\mathcal{L}(y) = \frac{1}{s+1}$$

$$s^2\mathcal{L}(y) - 0 - 1 + 5s\mathcal{L}(y) - 0 + 6\mathcal{L}(y) = \frac{1}{s+1} \quad \left[\begin{array}{l} \text{given} \\ y(0)=0 \\ y'(0)=1 \end{array} \right]$$

$$s^2\mathcal{L}(y) - 1 + 5s\mathcal{L}(y) + 6\mathcal{L}(y) = \frac{1}{s+1}$$

$$\mathcal{L}(y)[s^2 + 5s + 6] = \frac{1}{s+1} + 1 = \frac{1+s+1}{s+1} = \frac{s+2}{s+1}$$

$$\mathcal{L}(y)[s+3][s+2] = \frac{s+2}{s+1}$$

$$\mathcal{L}(y) = \frac{s+2}{(s+1)(s+2)(s+3)} = \frac{1}{(s+1)(s+3)} \quad \text{--- (1)}$$

$$\frac{1}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3} \quad \text{--- (2)}$$

$$1 = A(s+3) + B(s+1)$$

putting $s = -1$.

$$1 = A(-1+3) \Rightarrow 1 = 2A \Rightarrow A = 1/2.$$

putting $s = -3$

$$1 = B(-3+1) \Rightarrow 1 = B(-2) \Rightarrow B = -1/2.$$

substituting A, B in ②

$$\frac{1}{(s+1)(s+3)} = \frac{1/2}{s+1} - \frac{1/2}{s+2}.$$

$$\therefore \textcircled{1} \Rightarrow \mathcal{L}(y) = \frac{1/2}{s+1} - \frac{1/2}{s+2}$$
$$y = \frac{1}{2} \mathcal{L}^{-1}\left[\frac{1}{s+1}\right] - \frac{1}{2} \mathcal{L}^{-1}\left[\frac{1}{s+2}\right]$$

$$\underline{\underline{y = \frac{1}{2} e^{-t} - \frac{1}{2} e^{-2t}}}$$

6 Using Convolution theorem find the inverse Laplace Transform of $\frac{s^2}{(s^2+a^2)(s^2+b^2)}$

$$\text{let } F(s) = \frac{s}{s^2+a^2} \quad G(s) = \frac{s}{s^2+b^2}$$

$$f(t) = \mathcal{L}^{-1}(F(s)) = \mathcal{L}^{-1}\left[\frac{s}{s^2+a^2}\right] = \cos at$$

$$g(t) = \mathcal{L}^{-1}(G(s)) = \mathcal{L}^{-1}\left[\frac{s}{s^2+b^2}\right] = \cos bt$$

$$\begin{aligned}
 \mathcal{L}^{-1} \left[\frac{s^2}{(s^2+a^2)(s^2+b^2)} \right] &= \int_0^t f(u)g(t-u) du \\
 &= \int_0^t \cos au \cos b(t-u) du \\
 &= \int_0^t \cos au \cos(bt-bu) du
 \end{aligned}$$

$$\begin{aligned}
 \cos(A+B) &= \cos A \cos B - \sin A \sin B \\
 \cos(A-B) &= \cos A \cos B + \sin A \sin B
 \end{aligned}$$

$$\begin{aligned}
 2 \cos A \cos B &= \cos(A+B) + \cos(A-B) \\
 \cos A \cos B &= \frac{\cos(A+B) + \cos(A-B)}{2}
 \end{aligned}$$

$$\text{Here } A = au \quad B = bt - bu$$

$$\cos au \cos(bt-bu) = \frac{\cos(au+bt-bu) + \cos(au-(bt-bu))}{2}$$

$$= \frac{1}{2} [\cos(au+bt-bu) + \cos(au-bt+bu)]$$

$$\therefore \mathcal{L}^{-1} \left[\frac{s^2}{(s^2+a^2)(s^2+b^2)} \right] = \frac{1}{2} \int_0^t [\cos(au+bt-bu) + \cos(au-bt+bu)] du$$

$$= \frac{1}{2} \int_0^t \cos(au+bt-bu) du + \frac{1}{2} \int_0^t \cos(au-bt+bu) du$$

$$= \frac{1}{2} \left[\frac{\sin(au+bt-bu)}{a-b} \right]_0^t + \frac{1}{2} \left[\frac{\sin(au-bt+bu)}{a+b} \right]_0^t$$

$$= \frac{1}{2(a-b)} \left[\sin(a+b-t) - \sin b \right]$$

$$+ \frac{1}{2(a+b)} \left[\sin(a-b+t) - \sin(-t) \right]$$

$$= \frac{1}{2(a-b)} \left[\sin a \cos t - \sin b \cos t \right] + \frac{1}{2(a+b)} \left[\sin a \cos t + \sin b \cos t \right]$$

$$= \frac{1}{2} \left[\frac{\sin a \cos t - \sin b \cos t}{a-b} + \frac{\sin a \cos t + \sin b \cos t}{a+b} \right]$$

$$= \frac{1}{2} \left[\frac{(a+b)(\sin a \cos t - \sin b \cos t) + (a-b)(\sin a \cos t + \sin b \cos t)}{a^2 - b^2} \right]$$

$$= \frac{1}{2} \left[\frac{a \sin a \cos t - a \sin b \cos t + b \sin a \cos t - b \sin b \cos t + a \sin a \cos t + a \sin b \cos t - b \sin a \cos t - b \sin b \cos t}{a^2 - b^2} \right]$$

$$= \frac{1}{2} \left[\frac{2a \sin a \cos t - 2b \sin b \cos t}{a^2 - b^2} \right]$$

$$= \frac{a \sin a \cos t - b \sin b \cos t}{a^2 - b^2}$$

Find the inverse Laplace Transform of $\frac{s+8}{s^2+4s+5}$

$$\frac{s+8}{s^2+4s+5} = \frac{s+2+6}{s^2+4s+4+1} = \frac{s+2+6}{(s+2)^2+1}$$

$$= \frac{s+2}{(s+2)^2+1} + \frac{6}{(s+2)^2+1}$$

$$\mathcal{L}^{-1} \left[\frac{s+8}{s^2+4s+5} \right] = \mathcal{L}^{-1} \left[\frac{s+2}{(s+2)^2+1} \right] + 6 \mathcal{L}^{-1} \left[\frac{1}{(s+2)^2+1} \right]$$

$$= \underline{\underline{e^{-2t} \cos t + 6e^{-2t} \sin t}}$$

b) Using Laplace Transform solve.

$$y'' + 16y = 4 \delta(t-3\pi) \quad y(0)=2, \quad y'(0)=0.$$

$$y'' + 16y = 4 \delta(t-3\pi)$$

$$\mathcal{L}[y'' + 16y] = 4 \mathcal{L}[\delta(t-3\pi)]$$

$$\mathcal{L}[y''] + 16 \mathcal{L}[y] = 4 e^{-3\pi s}$$

$$[s^2 \mathcal{L}[y] - s y(0) - y'(0)] + 16 \mathcal{L}[y] = 4 e^{-3\pi s}$$

$$s^2 \mathcal{L}[y] - 2s - 0 + 16 \mathcal{L}[y] = 4 e^{-3\pi s}$$

$$\mathcal{L}(y) [s^2 + 16] = 4e^{-3\pi s} + 2s.$$

$$\mathcal{L}(y) = \frac{4e^{-3\pi s} + 2s}{s^2 + 16}.$$

$$y = 4 \cdot \mathcal{L}^{-1} \left[\frac{e^{-3\pi s}}{s^2 + 16} \right] + 2 \mathcal{L}^{-1} \left[\frac{s}{s^2 + 16} \right] \quad \text{--- (1)}$$

$$\frac{e^{-3\pi s}}{s^2 + 16} = e^{-as} F(s), \quad a = 3\pi \quad \text{where} \quad F(s) = \frac{1}{s^2 + 16} \Rightarrow f(t) = \frac{\sin 4t}{4}$$

$$\begin{aligned} \mathcal{L}^{-1} \left[\frac{e^{-3\pi s}}{s^2 + 16} \right] &= \mathcal{L}^{-1} \left[e^{-as} F(s) \right] = f(t - 3\pi) u(t - 3\pi) \\ &= \sin 4(t - 3\pi) u(t - 3\pi) \end{aligned}$$

substitute in (1) we get

$$y = \frac{4 \cdot \sin 4(t - 3\pi) u(t - 3\pi)}{4} + 2 \cos 4t.$$

$$y = \sin 4(t - 3\pi) u(t - 3\pi) + 2 \cos 4t.$$

Module 2

Find the Fourier Transform of $f(x) = \begin{cases} e^x, & \text{if } -a < x < a \\ 0 & \text{otherwise} \end{cases}$

Fourier transform of $f(x)$ is

$$\begin{aligned} \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^x e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{x(1-i\omega)} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{x(1-i\omega)}}{1-i\omega} \right]_{-a}^a \\ &= \frac{1}{\sqrt{2\pi} (1-i\omega)} \left[e^{a(1-i\omega)} - e^{-a(1-i\omega)} \right] \end{aligned}$$

b) Find the Fourier Cosine integral of $f(x) = \begin{cases} \cos x & \text{if } 0 < x < \pi/2 \\ 0 & \text{otherwise} \end{cases}$

Fourier Cosine integral of $f(x)$ is

$$f(x) = \int_0^{\infty} A(\omega) \cos \omega x \, d\omega.$$

To find $A(\omega)$:

$$\begin{aligned} A(\omega) &= \frac{2}{\pi} \int_0^{\infty} f(x) \cos \omega x \, dx. \\ &= \frac{2}{\pi} \int_0^{\pi/2} \cos x \cos \omega x \, dx. \quad \text{--- (i)} \end{aligned}$$

$$\begin{aligned} \cos x \cos \omega x &= \frac{\cos(x+\omega x) + \cos(x-\omega x)}{2} \\ &= \frac{\cos x(1+\omega) + \cos x(1-\omega)}{2} \end{aligned}$$

$$\text{(i)} \Rightarrow A(\omega) = \frac{2}{\pi \cdot 2} \int_0^{\pi/2} [\cos x(1+\omega) + \cos x(1-\omega)] \, dx$$

$$= \frac{1}{\pi} \left[\int_0^{\pi/2} \cos x(1+\omega) \, dx + \int_0^{\pi/2} \cos x(1-\omega) \, dx \right]$$

$$= \frac{1}{\pi} \left[\left[\frac{\sin x(1+\omega)}{1+\omega} \right]_0^{\pi/2} + \left[\frac{\sin x(1-\omega)}{1-\omega} \right]_0^{\pi/2} \right]$$

$$= \frac{1}{\pi(1+\omega)} \left[\sin \frac{\pi}{2}(1+\omega) \right] + \frac{1}{\pi(1-\omega)} \left[\sin \frac{\pi}{2}(1-\omega) \right]$$

$$= \frac{1}{\pi(1+\omega)} \left[\sin \left(\frac{\pi}{2} + \frac{\pi}{2}\omega \right) \right] + \frac{1}{\pi(1-\omega)} \left[\sin \left(\frac{\pi}{2} - \frac{\pi}{2}\omega \right) \right]$$

$$= \frac{1}{\pi(1+\omega)} \left[\cos \frac{\pi}{2}\omega \right] + \frac{1}{\pi(1-\omega)} \left[\cos \frac{\pi}{2}\omega \right]$$

$$= \frac{1}{\pi} \left[\frac{\cos \pi/2 \omega}{1+\omega} + \frac{\cos \pi/2 \omega}{1-\omega} \right]$$

$$= \frac{1}{\pi} \left[\frac{\cos \pi/2 \omega - \omega \cos \pi/2 \omega + \cos \pi/2 \omega + \omega \cos \pi/2 \omega}{1-\omega^2} \right]$$

$$= \frac{1}{\pi} \left[\frac{2 \cos \pi/2 \omega}{1-\omega^2} \right]$$

a) Find the Fourier Cosine Transform of

$$f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

) Fourier Cosine Transform of $f(x)$ is

$$f_c^{\wedge}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos \omega x dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 x^2 \cos \omega x dx \quad \text{--- (1)}$$

$$\int_0^1 x^2 \cos \omega x dx.$$

$$u = x^2$$

$$u' = 2x$$

$$u'' = 2$$

$$v = \cos \omega x$$

$$v_1 = \sin \omega x / \omega$$

$$v_2 = -\cos \omega x / \omega^2$$

$$v_3 = \frac{-\sin \omega x}{\omega^3}$$

$$\int_0^1 x^2 \cos \omega x dx = \left[x^2 \frac{\sin \omega x}{\omega} + \frac{2x \cos \omega x}{\omega^2} + \frac{2 \sin \omega x}{\omega^3} \right]_0^1$$

$$= \frac{\sin \omega}{\omega} + \frac{2 \cos \omega}{\omega^2} - \frac{2 \sin \omega}{\omega^3}$$

$$\therefore \textcircled{1} \Rightarrow \underline{\underline{f_c'(\omega) = \sqrt{\frac{2}{\pi}} \left[\frac{\sin \omega}{\omega} + \frac{2 \cos \omega}{\omega^2} - \frac{2 \sin \omega}{\omega^3} \right]}}$$

5) Find the Fourier Transform of $f(x)$

$$= \begin{cases} a - |x| & \text{if } |x| < a \\ 0 & \text{otherwise} \end{cases}$$

6) Fourier transform of $f(x)$ is

$$f'(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a f(x) e^{-i\omega x} dx$$

$$f(x) = a - |x|, \quad -a < x < a, \quad \begin{matrix} |x| = x & x > 0 \\ -x & x < 0 \end{matrix}$$

$$f(x) = \begin{cases} a - x & 0 < x < a \\ a + x & -a < x < 0 \end{cases}$$

$$f(x) = a+x \quad -a < x < 0.$$

$$= a-x \quad 0 < x < a.$$

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)

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-a}^0 (a+x) e^{-i\omega x} dx + \int_0^a (a-x) e^{-i\omega x} dx$$

(1)

consider $\int_{-a}^0 (a+x) e^{-i\omega x} dx.$

$$u = a+x$$

$$u' = 1$$

$$u'' = 0$$

$$v = e^{-i\omega x}$$

$$v_1 = \frac{e^{-i\omega x}}{-i\omega}$$

$$v_2 = \frac{e^{-i\omega x}}{(-i\omega)(-i\omega)} = \frac{e^{-i\omega x}}{-\omega^2}$$

$$\int_{-a}^0 (a+x) e^{-i\omega x} dx = \left[\frac{(a+x) e^{-i\omega x}}{-i\omega} + \frac{e^{-i\omega x}}{\omega^2} \right]_{-a}^0$$

$$= -\frac{1}{i\omega} [a] + \frac{1}{\omega^2} [1 - e^{i\omega a}]$$

(2)

Consider

$$\int_0^a (a-x) e^{-i\omega x} dx$$

$$u = a-x$$

$$u' = -1$$

$$v = e^{-i\omega x}$$

$$v_1 = \frac{e^{-i\omega x}}{-i\omega}$$

$$v_2 = \frac{e^{-i\omega x}}{(-i\omega)(-i\omega)} = \frac{e^{-i\omega x}}{-\omega^2}$$

$$\begin{aligned}
 \int_0^a (a-x) e^{-i\omega x} dx &= \left[\frac{(a-x) e^{-i\omega x}}{-i\omega} + \frac{e^{-i\omega x}}{-\omega^2} \right]_0^a \\
 &= \frac{-1}{i\omega} \left[0 - a e^0 \right] - \frac{1}{\omega^2} \left[e^{-i\omega a} - 1 \right] \\
 &= +\frac{a}{i\omega} - \frac{1}{\omega^2} e^{-i\omega a} + \frac{1}{\omega^2} \quad \text{--- (3)}
 \end{aligned}$$

substitute eq (2) and (3) in eq (1)

$$\begin{aligned}
 \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \left[\frac{-a}{i\omega} + \frac{1}{\omega^2} - \frac{e^{i\omega a}}{\omega^2} + \frac{a}{i\omega} - \frac{1}{\omega^2} e^{-i\omega a} + \frac{1}{\omega^2} \right] \\
 &= \frac{1}{\sqrt{2\pi}} \left[\frac{2}{\omega^2} - \frac{1}{\omega^2} (e^{i\omega a} + e^{-i\omega a}) \right] \\
 &= \frac{1}{\sqrt{2\pi}} \left[\frac{2}{\omega^2} - \frac{1}{\omega^2} 2\cos \omega a \right] \\
 &= \frac{2}{\sqrt{2\pi}} \left[-\frac{\cos \omega a}{\omega^2} + \frac{1}{\omega^2} \right] \\
 &= \sqrt{\frac{2}{\pi}} \left[-\frac{\cos \omega a}{\omega^2} + \frac{1}{\omega^2} \right]
 \end{aligned}$$